

Benjamin Poole

benjpoole01@gmail.com | benjaminpoole.github.io

A research oriented Computer Science graduate with First Class Honours. One of ten to be awarded the 2024 Google Europe Scholarship. Offered a funded PhD position at Southampton (skipping integrated masters) but seeking an industry career in Machine Learning and Informatics research. View my developer portfolio (strongly recommended) at benjaminpoole.github.io.

EDUCATION

University of Edinburgh

MSc Artificial Intelligence

Edinburgh, UK

September 2026 – Present

- Undertaking a Masters in Artificial Intelligence at the University of Edinburgh.

University of Southampton

BEng Computer Science

Southampton, UK

September 2023 – July 2026

- Graduated with First Class Honours, achieving 89% in my machine learning module and completing my [dissertation](#) in agents and reinforcement learning.

Cherwell School

A Levels & GCSEs

Oxford, UK

September 2015 – June 2022

- **A Levels:** Maths (A*), Further Maths (A), Physics (A).
- **GCSEs:** Maths (9), Physics (9), Computer Science (9), Biology (9), Geography (9), Chemistry (8), English Literature (8), English Language (7), Music (7), Citizenship (7).

I am also the 2024 recipient of the [Google Europe Scholarship](#) for students with disabilities.

RESEARCH EXPERIENCE

Researcher

March – September 2024

Agents Interaction and Complexity research group - University of Southampton

Researched the development of a vision language policy to control robot agents, supervised by a Postgraduate researcher.

- Developed a novel idea based on the MineCLIP dataset ([arxiv:2206.08853](#)) to build a teacher model that could train a student model using Imitation Learning.
- In my 2nd year I was invited back to research swam robotics for [XPRIZE Wildfires](#) but the university's funding was cancelled. I have now been offered a funded PhD with this research group.

Researcher

October 2022 – August 2023

Tupel 3D

Researched and prototyped a low-cost, structured light hand-held 3D scanner.

- Self-taught computer vision and SLAM reconstruction from open online lectures.
- Developed a proprietary AI to create a 3D depth reconstruction system that could provide accurate 3D imagery using only primitive hardware. The company believe that this system is unique and superior to all publications they know.
- Developed a unique SLAM camera tracking system following ORB-SLAM 3 ([arxiv:2007.11898](#)) that would allow fast camera tracking for 3D reconstruction.

ENGINEERING EXPERIENCE

Software Engineer

August – September 2025

Britwind Ecotricity

- Developed an IoT energy dashboard using web technologies, to display customer wind turbine power generation allowing full analytics and customisation in 6 weeks. This saved significant development costs by replacing a previously stalled, contractor-dependent effort.

FEATURED PROJECTS

I enjoy self-teaching ML and theoretical computer science. To support my learning I have undertaken many personal projects, demonstrated interactively on [my website](#).

Ferranti Mark 1 Assembler and Emulator (2025–26)

[Interactive demo](#)

- Built a browser-based emulator of the 1951 Ferranti Mark 1, with an assembler from a small assembly language to the machine's native code, including its five-bit teleprinter encoding, with live visualisation of the Williams-tube electronic store.

From-scratch implementation and analysis of an LLM (2024)

[Write-up](#)

- Developed a 14.5m parameter language model from scratch (matrix multiplication), following the TinyStories paper ([arXiv:2305.07759](#)). The model was able to coherently model story language despite its small size.
- Analysed the model word vectors using 2D T-SNE dimensionality reduction to understand how the model interprets word semantics in terms of their context and relation to other words.
- Reading of academic papers to understand how LLM technologies work such as A Survey of Large Language Models ([arxiv:2303.18223](#)) and Locating and Editing Factual Associations in GPT ([arxiv:2202.05262](#)) which taught me LLM interpretability analysis and how LLMs store and retrieve facts.

Implementing a diffusion model to convert paint sketches to photorealistic images (2023) [Write-up](#)

- An implementation of the paper SDEdit ([arxiv:2108.01073](#)) to learn more about image prompting for generative AI.
- This involved experimenting with a diffusion model, following [arxiv:2112.10752](#) and then modifying an implementation of Stable Diffusion to add image prompting capability such that the model could transform a simple MS Paint sketch to a photorealistic image.

MORE PROJECTS

Browser-native speech transcriber (2026)

[Interactive demo](#)

- NVIDIA Parakeet TDT 0.6B speech recognition running entirely in the browser, with no server.

Ecumenical Christian Library (2025)

[Live site](#)

- A searchable digital library of over 17,000 texts.

Compiler for a C-like language (2025)

- Lexer, parser and assembly code generator, written as a self-directed project following summer reading on programming language theory: type systems, static analysis and interactive theorem proving.

Fast Mandelbrot renderer

[GitHub](#)

- A GPU-accelerated Mandelbrot set explorer built for speed.

Pre-university projects

- CNN model to drive a virtual car in a game environment.
- Using GANs and Autoencoders to generate human faces.
- Building a neural network from scratch using matrix algebra in NumPy, to further my understanding of the mathematical basis of AI.

OTHER INTERESTS

While my primary hobby is in computer science I also maintain an interest in electronics and philosophy.

- Twice winner of school science fair with audio electronics.
- Developed a solid-state Tesla Coil that can reach 60kV.
- ZVS (Zero Voltage Switching) induction heater from scratch.
- Built a 4 bit adder from 74xx series logic chips (with a friend) to demonstrate the workings of a CPU's ALU.
- I enjoy playing the drums and astrophotography: I have imaged the gravitationally lensed quasar (Q0957+561) 8.7 billion light years away using a DSLR camera with a tracking mount.